Learning Task 2

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Independence

Definition 2.1: Independence

Two events A and B are *independent* if:

$$P(A|B) = P(A) \quad \text{or} \quad P(B|A) = P(B)$$

Corollary 2.2: Product Rule for Independent Events

If A and B are independent, then:

$$P(A \cap B) = P(A) \cdot P(B)$$

This follows immediately from probability rule (R4.1)[cite: 2].

Theorem 2.3: Independence of Complements

If A and B are independent, then their complements A' and B' are also independent[cite: 2].

Proof

Using De Morgan's Law and probability rules:

$$\begin{aligned} P(A' \cap B') &= P((A \cup B)') \\ &= 1 - P(A \cup B) \\ &= 1 - P(A) - P(B) + P(A \cap B) \\ &= 1 - P(A) - P(B) + P(A) \cdot P(B) \quad \text{(by independence)} \\ &= (1 - P(A)) \cdot 1 - (1 - P(A)) \cdot P(B) \\ &= (1 - P(A))(1 - P(B)) \\ &= P(A') \cdot P(B') \end{aligned}$$

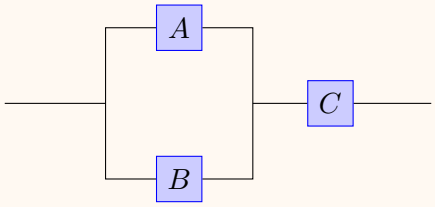
Example 2.4: Dependence in semiconductor manufacturing

Consider the semiconductor manufacturing scenario from the previous learning task:

$$\left.\begin{aligned} P(B) &= 0.024 \\ P(B|A) &= 0.10 \end{aligned}\right\} \implies A \text{ and } B \text{ are dependent events.}$$

Example 2.5: Reliability of a three-device circuit

Given a circuit with three components A, B, and C, assume that the individual devices function independently. Let $P(A) = 0.95$, $P(B) = 0.95$, and $P(C) = 0.90$ denote the operational probabilities.



Figuur 1: Circuit diagram with three independent devices.

Let the individual operational probabilities be given as $P(A) = 0.95$, $P(B) = 0.95$, and $P(C) = 0.90$. The overall probability that the circuit functions is formulated as:

$$P(\text{circuit functions}) = P((A \cup B) \cap C) = P(A \cup B) \cdot P(C)$$

To compute the parallel subsystem reliability $P(A \cup B)$:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - P(A) \cdot P(B) \\ &= 0.95 + 0.95 - (0.95)^2 \\ &= 0.9975 \end{aligned}$$

Thus, the operational probability is:

$$P(\text{circuit functions}) = 0.9975 \cdot 0.90 = \mathbf{0.89775} \quad (\approx 0.8978)$$

Alternative method for parallel components An alternative approach to evaluating the parallel subsystem $A \cup B$ uses complement probabilities:

$$\begin{aligned} P(A \cup B) &= 1 - P((A \cup B)') \\ &= 1 - P(A' \cap B') \\ &= 1 - P(A') \cdot P(B') \quad \text{(by independence)} \\ &= 1 - (1 - P(A))(1 - P(B)) \\ &= 1 - (0.05)^2 = 0.9975 \end{aligned}$$

So $P(\text{circuit functions}) = 0.9975 \cdot 0.9 = \mathbf{0.89775}$.

Bayes' Theorem

We revisit once again the semiconductor example from the previous learning task. Suppose a semiconductor fails (B). To find the probability that the chip was subjected to high contamination levels during manufacturing (A), we evaluate the inverse conditional probability $P(A|B)$.

When evaluating conditional probabilities in reverse operational order, Bayes' Theorem is applied.

Theorem 2.6: Bayes' Theorem

For two events A and B with $P(B) > 0$, it holds that:

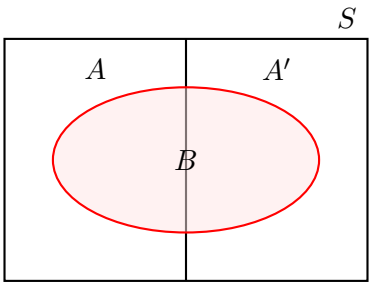
$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}.$$

Proof of Bayes' Theorem

From rules (R4.1) and (R4.2), we can express $P(A \cap B)$ both as $P(B|A) \cdot P(A)$ and $P(A|B) \cdot P(B)$.

Dividing both expressions by $P(B)$ (which is allowed, as $P(B)$ is not zero) yields:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A) \cdot P(A)}{P(B)}$$



Figuur 2: Partitioning of event B across sample space partitions A and A'.

By definition of conditional probability and the Law of Total Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

Example 2.7: Semiconductor manufacturing

Revisiting the semiconductor example, we calculate $P(\text{high contamination} \mid \text{semiconductor fails}) = P(A|B)$:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B|A) \cdot P(A) + P(B|A') \cdot P(A')} = \frac{0.10 \cdot 0.20}{0.024} = \frac{0.02}{0.024} \approx \mathbf{0.8333}$$

Random Variables

Given a random experiment E with associated sample space S (for instance, engaging a target in up to five attempts, $S = \{h, mh, mmh, mmmh, mmmmh\}$ where $m =$ miss and $h =$ hit):

Definition 2.8: Random Variable

A function X that assigns a real number to each outcome in the sample space S is called a *random variable*.

- A **discrete random variable** is a random variable with a finite or countably infinite range.
- A **continuous random variable** has a range consisting of an interval (either finite or infinite) of real numbers.

Example 2.9: target engagement

In engaging a target up to five attempts, let X represent the attempt number on which the target is eliminated. Here, $X(mmh) = 3$ and $X(h) = 1$. Because the range of X is the finite set $\{1, 2, 3, 4, 5\}$, X is a discrete random variable.

Example 2.10: lifespan analysis

Let X denote the operational lifespan of a light bulb. Since the range of $X = \{x \in \mathbb{R} \mid x \geq 0\}$ forms a continuous real interval, X is a continuous random variable.

Example 2.11: keep trying until first success

Consider engaging a target in an infinite sequence of independent attempts with a constant single-trial success probability p. Let X denote the attempt number on which the target is eli-

minated. Because the range of $X = \{1, 2, 3, \dots\}$ is countably infinite, X is a discrete random variable.